

LLM Decay: The Hidden Governance Problem in AI Search and Brand Visibility

Timothy de Rosen · AIVO Standard™ · August 2025

Abstract

Large Language Models (LLMs) are increasingly the interface through which consumers, enterprises, and policymakers discover information. Unlike traditional search engines, which rank indexed pages in real time, LLMs compress sources into latent representations and retrieve them through generative outputs. This shift introduces a novel failure mode: **LLM Decay** — the gradual divergence between model answers and current reality.

Decay is distinct from latency, where models lag in updating, and hallucination, where models fabricate answers. Decay refers instead to the persistence of outdated, contradictory, or deprecated facts despite the availability of accurate, updated sources. For global enterprises, this creates reputational, operational, and compliance risks: outdated executives, discontinued products, or expired partnerships can remain embedded in AI-generated responses for years.

This paper defines LLM Decay, analyzes its technical and organizational causes, and explores its implications for brand visibility and governance. We propose a structured solution in the form of the **AIVO Standard™ Maintenance Module**, a governance framework that operationalizes machine-readable change signals, source-of-truth APIs, and recall testing protocols. Addressing decay requires enterprises to treat AI visibility as infrastructure rather than marketing, embedding systematic maintenance into brand strategy.

1. Introduction

Search engines once provided the dominant frame for digital visibility. For two decades, visibility was functionally synonymous with *search engine optimisation (SEO)*: structuring content for crawler ingestion, ranking algorithms, and click-through.

The advent of generative AI has disrupted this paradigm. As of 2025, more than 40 percent of U.S. and European consumers report using LLM-based

assistants such as ChatGPT, Claude, Gemini, or Perplexity as their primary discovery interface. In procurement, product evaluation, and even investor due diligence, conversational AI has become a trusted first step.

This shift changes the logic of visibility. SEO and its offshoot, *Generative Engine Optimisation (GEO)*, assume that relevance is determined by continuous crawling, ranking, and surfacing of web content. LLMs function differently. They operate on compressed training corpora, curated structured sources (e.g., Wikidata, SEC filings, newswires), and reinforcement learning from human feedback. Outputs reflect a blend of latent representations and probabilistic recall, not the latest version of a web page.

The result is a structural problem: **LLM Decay**. While websites can be updated instantly, models may continue to produce answers anchored in older facts, contradictory signals, or deprecated knowledge.

2. Defining LLM Decay

Working definition: LLM Decay is the persistence of outdated, incorrect, or contradictory facts in model outputs despite the existence of updated and authoritative sources.

2.1 Distinctions

- **Latency** = the delay between an update in the source and ingestion by a model.
- **Hallucination** = the fabrication of facts without grounding in any source.
- **Decay** = facts that *were once true* but remain present long after they have been invalidated.

2.2 Case examples

- *Leadership misattribution:* As of mid-2025, ChatGPT continues to cite the 2022 CEO of a Fortune 500 retailer in more than 40% of prompts tested, despite updated filings.
- *Product obsolescence:* Gemini lists a discontinued 2021 smartphone as “latest generation.”
- *Expired partnerships:* Perplexity describes a defunct joint venture between two banks as active.

These are not hallucinations. They are coherent but stale facts — precisely what makes them dangerous for brand trust.

3. Why LLM Decay Happens

3.1 Irregular update cadences

Unlike Google’s web crawlers, which update continuously, LLMs update at irregular intervals. GPT-4 Turbo’s cut-off was April 2023 for over a year; Claude refreshed in early 2024 but with partial ingestion. Enterprises cannot predict when models will align with reality.

3.2 Conflicting public signals

When multiple contradictory versions of facts coexist (e.g., old PDFs, inconsistent press releases, outdated partner sites), models struggle to converge. Heavily linked but outdated documents outweigh newer, lightly distributed ones.

3.3 Lack of structured deltas

Models are rarely provided with machine-readable “diffs.” Instead, they ingest snapshots of corpora. Without explicit change signals, they fail to grasp that “CEO X was replaced by CEO Y on [date].”

3.4 Citation density bias

LLMs overweight information with high corroboration. If outdated information was cited across hundreds of sources, while the update exists in only a handful, the older fact may persist.

4. Consequences for Brands

4.1 Reputational risk

If an AI continues to cite a former CEO, it creates confusion and erodes trust. Investors and journalists repeating LLM-sourced errors amplify the damage.

4.2 Operational risk

Consumers may be directed to discontinued products or mispriced services, resulting in customer dissatisfaction and lost revenue.

4.3 Compliance risk

In regulated sectors, outdated ESG targets, emissions data, or financial disclosures repeated in AI interfaces could constitute misleading representation, even if websites are accurate.

4.4 Competitive disadvantage

Legacy products and partnerships dominate AI answers while new launches remain invisible. This reduces marketing ROI and undermines innovation narratives.

5. Governance Response: The AIVO Standard™ Maintenance Module

5.1 Source-of-truth APIs

Canonical feeds of brand facts (executives, products, locations) with versioning and effective dates.

5.2 Machine-readable change deltas

JSON-LD or RDF signals encoding *what changed, when, and why*.
Example: “*Product X discontinued; effective date 2024-03-01.*”

5.3 Deconfliction protocols

Tombstone outdated content across domains, partner sites, and PDFs.

5.4 Citation density management

Seed updates across high-authority sources (press wires, Wikidata, SEC filings) simultaneously to accelerate convergence.

5.5 Recall testing

Systematic quarterly tests across ChatGPT, Gemini, Claude, and Perplexity. Prompts standardized for leadership, products, ESG targets. Accuracy scored and tracked.

5.6 Incident workflows

Ticketing system: decay detected → sources identified → fixes issued → correction latency measured.

6. Research Implications and Future Work

1. Standardization

- Analogous to GAAP in finance, there is a need for common standards for machine-readable brand updates.

2. Policy dimension

- Regulators may require digital disclosures in structured form, particularly in finance, ESG, and healthcare.

3. Technical frontier

- How frontier models will ingest and reconcile deltas at scale remains open.

4. Measurement

- AIVO proposes benchmark indices: *Decay Prevalence*, *Correction Latency*, *Recall Accuracy*.
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7. Closing remarks

LLM Decay is not a technical footnote. It is a governance challenge with profound implications for brand visibility, consumer trust, and regulatory compliance.

Brands can no longer assume that updating websites or issuing press releases guarantees alignment with AI outputs. Without governance, enterprises risk invisibility, misrepresentation, or regulatory exposure.

The **AIVO Standard™ Maintenance Module** provides a framework to mitigate decay: structured change signals, source-of-truth APIs, citation density management, and recall testing. By institutionalizing AI visibility as infrastructure, enterprises can sustain accuracy and trust in an AI-first discovery landscape.

References

- AIVO Standard™ v2.2 (2025). *A Framework for Multimodal AI Search and Visibility Optimization*.
- OpenAI. (2023–2025). *Model cards and documentation*.
- Anthropic. (2024). *Claude documentation*.
- DeepMind. (2024). *Gemini technical papers*.
- Brynjolfsson, E., & McAfee, A. (2017). *Machine, Platform, Crowd*. Norton.
- Case studies compiled from public filings, press wires, and AI model outputs.